GRAYCO
CONSTRUCTION
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GRAYCO STUDY GUIDE

Water Damage Restoration Key Terms

Every key term from the course with a plain-English definition, grouped by subject. Use it to review before the GrayCo test and the IICRC WRT exam.

98 KEY TERMS | **9** SUBJECTS

GRAYCO CONSTRUCTION SERVICES, LLC · BARTLETT, TENNESSEE

STUDY GUIDE

Key terms and definitions

01 Safety

Hazard assessment
Do a site-specific hazard assessment on every job; it is the first priority of the initial inspection. You have to know a hazard exists before you can protect yourself.

Slips, trips and falls
The most common hazards on a water job.

General Duty Clause
OSHA rule that an employer must keep the workplace free of recognized hazards. OSHA also requires a written respiratory protection program.

SDS
Safety data sheet. Required for every chemical, kept where it is stored, on the job, in the truck, and available to clients. Best product information: SDS, label and training.

Proper dilution
Mix products with the right amount of water; too much or too little wastes product, adds hazard or damages materials.

Hierarchy of controls
First eliminate the hazard, then engineer it out; PPE is the last resort.

One size does not fit all
Different hazards need different suits, cartridges and gloves; cheap PPE is usually the worst PPE.

Respirators
Medical evaluation and a fit test are required before wearing one.

Sewage PPE
PPE must protect all potential routes of exposure. In Category 3, the health of workers and occupants comes first.

Primary health care provider (PHCP)
Employees should see their doctor about vaccines (Hep B, tetanus). Hep B is three injections; after a suspected exposure, treatment within 24 hours.

Chemically sensitive customers
Should stay out of affected areas until work is done; sensitive or high-risk occupants (pregnant, infants, elderly, chemo, immunocompromised, recent surgery) return on their doctor's advice.

Confined spaces
Attics, crawl spaces, pits, etc. Always two people on site; permit-required spaces have extra OSHA rules.

Lead and asbestos
Regulated hazards that require compliance with federal law.

Wastewater disposal
Even clean-water extraction water is disposed of following all laws and regulations.

02 Tools and Equipment

Moisture sensor
Beeps/blinks; only for finding high moisture in carpet and pad and the edge of migration. Don't use it on other materials.

Moisture meters
Give a numerical value. Two types: penetrating (invasive) and non-penetrating (non-invasive).

Non-invasive meter
Radio frequency or conductive pads; no damage; good for baselines and mapping. Metal behind the surface causes false positives.

Invasive (pin) meter
Measures conductivity between pins; gives a reading at the pin depth.

Thermo-hygrometer
Measures air temperature and relative humidity; readings are recorded daily.

IR camera
Shows warmer and cooler, not wet and dry. Confirm with a moisture meter.

Air movers
Largest part of the inventory; blend air so wet materials evaporate into drier air.

Dehumidifiers
Two types: refrigerant and desiccant. They pull vapor out of the air, creating a 'thirst'.

Conventional refrigerant
Removes water by condensation on cold coils; struggles below about 55 GPP.

Low grain refrigerant (LGR)
Also condensation, with a pre-cooler; far more efficient and pulls air down to about 34 GPP.

Desiccant dehumidifier
Chemical attraction (adsorption), not condensation; not limited by dew point; makes the lowest vapor pressure. Check power requirements.

Air filtration devices (AFDs)
Air scrubbers; control dust, spores and contaminants; consider based on category of water.

Extraction equipment
Truck mounts, portables, extractors. Extraction is paramount.

Inner-air-cavity drying system
Pushes warm dry air into cavities: under cabinets, hardwood floors, behind vinyl wallpaper.

IICRC S500
The standard of care for professional water damage restoration.

03 New Claim

Work authorization (WA)
The contract: who pays and how. No work starts until it is signed.

Set expectations
Explain what the customer will see, hear and smell and the order of work.

Initial inspection
Stop the source first, then find what is wet, how wet, whether it is damaged and whether it is salvageable. Inspect up, down and horizontally to find the edge of migration.

Protect contents
Blocks and tabs under furniture, lift drapes and clothing, raise metal items.

Dry standard / drying goal
Average readings on unaffected like materials = dry standard. Goal is within 10% of it (S500). Equipment stays until goals are met.

Document preexisting damage
Before starting, and communicate it to all materially interested parties.

Extraction
The most effective way to speed drying. Primary extraction stops further migration.

Primary vs. secondary damage
Primary: damage from direct contact with the water. Secondary: damage from the wet environment left behind (humidity, mold).

04 Categories of Water

Category 1
From a sanitary source; no substantial risk from skin contact, ingestion or inhalation. Time and dirty materials can make it worse.

Category 2
Significant chemical, biological or physical contamination; can cause sickness or discomfort (fish tanks, toilet overflow with urine, washer/dishwasher discharge).

Category 3
Grossly unsanitary or carries pathogens (formerly 'black water'): sewage backflow from beyond the trap, and surface/ground water such as overflowing rivers.

Category 3 procedures
Remove highly porous materials; carpet and pad saturated with Cat 3 are removed (Cat 2: pad only); clean surfaces; apply biocide. If the client refuses proper procedures, stop work; waivers do not protect you.

No Category 4
Mold and regulated hazardous materials are 'special situations'. Drinking is not the test for category.

05 Class of Loss

Class of loss
Estimates the evaporation load; one of the inputs for equipment calculations. Based on the percentage of wet porous surface (ceiling + walls + floor).

Class 1
Least evaporation: wet porous materials under 5% of combined surfaces.

Class 2
Significant evaporation: 5%–40%; wicking under 24"; carpet and pad in place.

Class 3
Greatest evaporation: more than 40%; often from overhead; moisture above 24" on walls. Overhead water alone doesn't make it Class 3 or Cat 3.

Class 4
Specialty drying: deeply held water in low-porosity materials (hardwood, plaster, brick, concrete, stone, multiple layers). Needs LGR or desiccant and low vapor pressure.

06 Principles of Drying

Three ways to remove water
Extraction (liquid; the most effective), evaporation, and dehumidification/ventilation.

Psychrometry
The science of measuring air and moisture; used to track progress and make decisions on every job.

H.A.T. (the drying pie)
Humidity, Airflow and Temperature control the rate of evaporation.

Specific humidity
The actual amount of water vapor in the air: GPP (humidity ratio), vapor pressure, dew point.

Relative humidity (RH)
Moisture in the air as a % of the most it could hold at that temperature; 100% = saturated. Warmer air: RH goes down; cooler: RH goes up.

Hygroscopic materials
Materials that take up or hold moisture; their equilibrium moisture content (EMC) depends on RH.

GPP (humidity ratio)
Grains of moisture per pound of dry air. GPP, vapor pressure and dew point rise and fall together.

Airflow
Breaks the boundary layer of saturated air at the surface.

Temperature / energy
Warming a wet material adds energy and raises the vapor pressure of its water, speeding evaporation.

Vapor pressure
High pressure moves toward low pressure until equilibrium. Drives moisture from wet materials into drier air.

Evaporation
Liquid to vapor (energy added); the surface gets cooler.

Condensation
Vapor to liquid (energy removed); surfaces below the dew point condense water.

Sublimation
Solid to vapor without becoming liquid; freeze-drying books and documents.

Deposition / melting / freezing
Vapor to solid; solid to liquid; liquid to solid.

Dew point
Temperature at which air is saturated (100% RH). 75°F / 60% RH has a dew point around 62°F. Over 60% RH risks secondary damage.

Permeance
How easily vapor passes through a material. Under 1.0 perm = vapor retarder/barrier (vinyl, poly, some paints).

07 Calculations

Open / closed / combination drying
Open uses outside air; closed uses only mechanical dehumidification (use it when outside GPP is higher); combination uses both.

AHAM rating
Refrigerant dehumidifiers are rated in pints per day at 80°F / 60% RH.

Desiccant rating
Rated in CFM (cubic feet per minute), not ACH.

Dehu factors
Conventional: Class 1 = 100, 2 = 40, 3 = 30, 4 = n/a. LGR: 100 / 50 / 40 / 40. Desiccant ACH: 1 / 2 / 3 / 3.

Refrigerant formula
Cubic feet ÷ factor = pints per day; ÷ unit AHAM rating = number of units (round up).

Desiccant formula
Cubic feet × ACH ÷ 60 = CFM; ÷ unit CFM = number of units (round up). Also works for AFDs.

Airmover formula
1 per wet room + 1 per 50–70 SF wet floor + 1 per 100–150 SF wet wall/ceiling above 2 ft + 1 per inset/offset over 18". Round up. Limited migration: 1 per 14 LF of wet wall.

Air mover placement
Calculated room by room; all facing the same direction, never opposing.

Class 4 airflow
After surface water evaporates, reduce airflow velocity and increase vapor pressure differential (heat, drier air).

Types of air movers
Carpet driers (high velocity), low-pressure axial (volume, low amps), high-pressure axial (ductwork), inner-air systems.

Standing water
L × W × depth (ft) = cubic feet; × 7.48 = gallons; × 8.34 = pounds. Inches ÷ 12 = feet.

Pumping basements
Don't pump too fast: saturated soil outside presses on the walls and can collapse the foundation. Gallons ÷ pump flow rate = pumping time; if water keeps coming in, stop and get an engineer.

08 Structural Components

Carpet and pad by category
Cat 3: remove both. Cat 2: discard pad; extract, clean, dry carpet. Cat 1: consider in-place drying.

In-place drying test
Only if all yes: Cat 1, under 72 hours, deep extraction tools on hand, and LGR or desiccant dehumidification available.

Knee kicker and awl
Used to float carpet and remove wet pad when deep extraction isn't possible.

Partial floating of carpet
Carpet acts as a tent to dry both sides.

Vinyl over wet underlayment
Vinyl is a vapor barrier; usually removed.

Concrete
Very resistant itself; problems come when its moisture reaches other materials.

Hardwood floors dry
Within 2–4 percentage points of EMC; subfloor within 2–4 points of the hardwood. Probe in the dark (winter) grain.

Cupping
Edges up; bottom wetter. Usually dries out.

Crowning
Center up; top wetter (convex cupping).

Buckling
Boards lift off the subfloor; usually must be replaced.

Engineered wood
Plywood holds up best; OSB; particle board usually replaced; MDF suffers the most.

Drywall
Recovers well and is stronger once dry, but sagging ceilings are a hazard: drain and remove. Paper face molds if not dried fast.

Wall finishes
Flat paint dries easily; vinyl wallpaper is a vapor barrier; may need to drill holes under the base for cavity airflow.

Insulation
Wet cellulose is removed; Cat 2/3 insulation is removed; check for compacted batts.

Daily monitoring
At least every 24 hours, same tech and same meters in the same spots.

09 Microbial Issues

Antimicrobials
Use only when appropriate and follow the label. Wear PPE, have occupants leave and remove pets before applying.

Sanitizer
Reduces pathogens to a safe level; lowest level of biocidal activity.

Disinfectant
Destroys or inhibits microorganisms but not necessarily spores.

Sterilizer
Destroys all microorganisms and their spores.

Microbial growth needs
Spores, organic food, moisture and still air. Remove any one and growth stops. MVOCs are the odor.

Odors
Musty = mold; sharp/bitter = bacteria. Notify your supervisor calmly; wear PPE.

Third party experts (IEP)
Indoor Environmental Professional for extensive contamination, Cat 3 and high-risk situations.

Fungi and wood
Surface mold ~16% MC; dry rot 20–29%; wet rot 30%+. Mold can produce mycotoxins.